



The IoT For Environmental Monitoring: Needs And Challenges In India

This article is based on a speech given by Poonam J. Prasad, senior scientist, Analytical Instrumental Division, CSIR - National Environmental Engineering Research Institute, Nagpur, at IOTSHOW.IN 2019, held in Bengaluru. The institute's Analytical Instrumental Division focuses on R&D on environmental sensors and the Internet of Things (IoT). Speaking on the use of the IoT for environmental monitoring, Prasad highlighted the need for IoT-based environmental sensors to conserve energy, water and other resources



We need the Internet of Things (IoT) in environment monitoring to be able to conserve energy, water and other natural resources, which are being contaminated every second. In conventional environmental monitoring methods, samples are collected, analysed and analytical instrumentation is carried out on them.

There are two ways of doing this. One is manual, where the sample

is collected and analysed in a lab. Second is instrumental, where the quantity of pollutants in the sample is analysed, on the go, automatically.

Instrumental methods have direct analytics, where readings and results are automatically received. Manual methods, on the other hand, need pre-treating the sample before carrying out sedimentation, isolation and other processes on it.

When we talk about environ-

mental monitoring using the IoT, we primarily focus on such areas as waste management, air pollution and extreme weather.

Why we need the IoT for environment monitoring

When we go deep into environment monitoring, it is a very complex system and, hence, we cannot just start using sensors for regulatory purposes. If we have data for water and air, then we can use AI and ML tools, among others. There are environmental sensors for measuring water quality, radiations and hazardous chemicals.

Similarly, in the industrial IoT (IIoT), we need methods for ensuring safety of workers, because some industries generate obnoxious gases like sulphur, methane and sulphur's compounds, which are bad for human health. By getting data out of sensors, we can maintain a good safety record. Places that are inaccessible can also effectively utilise sensors.

Since 2012, research is happening all over the world on environmental sensors. Some reviews have already been done. People have done outdoor air-quality monitoring using a ZigBee-based wireless sensor network. However, indoor environ-